

Comments on Yalcin

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1. Some ways of connecting ‘if’ to conditional credence

Ramsey’s Thesis (schematic form): Whether one believes that if A, then C is a matter of one’s conditional credence that C given that A.

Canonical interpretation (in effect replacing ‘whether’ with ‘how strongly’):

Stalnaker’s Thesis: Whenever sentences ϕ and ψ express propositions p_1 and p_2 in a context, ‘If ϕ , ψ ’ expresses in that context a proposition q such that necessarily, any [rational?] person’s credence in q equals their conditional credence in p_2 given p_1 (assuming this is well-defined).

This turns out to have absurd consequences. One response is to say that conditionals don’t express propositions at all. But what is the alternative way of making a link between conditionals and conditional credence? One tradition focuses on:

Adams’s Thesis: Whenever sentences ϕ and ψ express propositions p_1 and p_2 in a context, the degree of assertability of ‘If ϕ , ψ ’ in that context for a [rational?] person is equal to that person’s conditional credence in p_2 given p_1 (if well-defined).

But the notion of “degree of assertability” here is deeply obscure. A more helpful thing to focus on, which people in this tradition also like, is:

Stalnaker’s Schema: Necessarily, any [rational?] person’s degree of confidence that if A, C equals their conditional credence that C given that A (if well-defined).

Note that this doesn’t talk about propositions at all, so it’s consistent with the view that conditionals don’t express propositions. Although it’s stated using the technical concept of “conditional credence”, it’s intended to extend to ordinary (non-numerical) ways of talking about degrees of confidence. For example: To be *almost sure* that if A, C is to be *almost as sure* that A and C as that A.

OK, but how does this extend to other kinds of attitude reports, such as belief reports? There’s a venerable idea that connects the verbs ‘believe’ and ‘think’ with the gradable adjective ‘confident’:

Lockeanism: On every context/precisification, there is a threshold t such that ‘believes/thinks’ is semantically equivalent to ‘has a degree of confidence greater than t ’. “To think that A is to be sufficiently confident that A.”

- Where should Lockeanism locate the threshold? Traditionally it’s thought to be typically pretty high (say > 0.8), though lower than the thresholds for ‘sure’ and ‘certain’. But there are examples that suggest that it can be quite low, maybe even < 0.5 . Being

a Lockean who allows for lowish thresholds is one way (but not the only way) of thinking that 'belief is weak'.

Putting *Stalnaker's Schema* together with *Lockeanism*, we get this, which Seth tentatively endorses:

Conditional Belief Schema: There is a threshold t such that to believe that if A, C is to for one's degree of confidence that C given that A to be greater than t .

2. Some counterexamples to Stalnaker's Schema and the Conditional Belief Schema

Scenario: we know that John's in a burning building. He's not suicidal so he's incredibly unlikely to jump without a net in place, instead he'd risk the stairs. We just found out that the fire engine is stuck in traffic and so that it's unlikely to make it in time. The following speeches look pretty good here:

- (1) It's beginning to look like he'll die if he jumps.
- (2) I'm increasingly confident that he'll die if he jumps.
- (3) I think he'll die if he jumps.

Nevertheless, our conditional credence that John's going to die given that he jumps is still very low.

- I take the moral of these cases to be that conditionals (and other modals) have a contextual parameter that can be filled by a partition. This could be added to many kinds of semantics.

3. Relating Lockeanism to expressivism about 'likely'

Expressivism about 'likely': For some t , to believe that it is likely that A is for one's degree of confidence that A to be greater than t .

If you like both this and Lockeanism, the question arises whether it's the same threshold or a different one. If it's the same, we'll get a surprising equation:

Believing Is Believing Likely: To believe that A is to believe that it is likely that A.

- This has the surprising consequence that to believe that A is to believe that it is likely to be likely that A, and so on. But this may be fine. The semantics for 'likely' that Seth likes already makes 'it's likely that A' semantically equivalent to 'it's likely that it's likely that A'.
- Worry: 'Not-P although he believed P' entails 'He believed something false' / 'He was wrong', but 'Not-P although he believed it was likely that P' doesn't. About this sort of worry, Seth says that 'it might be okay if "A believes that ϕ " is subject to certain norms of evaluation that "A believes that probably ϕ " isn't [subject to]. (Is that describing a way of rejecting *Believing Is Believing Likely*, or an addition to it?)

- Personally, I don't see much pressure on the Lockean who's also an expressivist about 'likely' to think that the two thresholds are required to be the same.

How do we modify standard dynamic/domain semantic treatments of 'likely', 'might', 'must' to make them Lockeanism-friendly (or to make sense of ascriptions of confidence involving them)?

Beddor and Goldstein's proposal (redone in a domain-semantic idiom, where sentences have truth values relative to pairs of worlds and information-states):

$$\llbracket A \text{ believes } \phi \rrbracket^{w,s} = 1 \text{ iff } A\text{'s credence at } w \text{ in } \{w': \llbracket \phi \rrbracket^{w', I(w,A)} = 1\} \text{ is above } t$$

where $I(w,A)$ is A 's information-state at w . Similarly for 'A is confident/sure/certain that ϕ ', with different thresholds.

- *Worry:* this makes 'A believes must ϕ ' equivalent to 'A is certain that must ϕ '. But consider: 'That's Trump, that's Ivanka, that's Donald Jr... I think that must be Donald Trump, Jr., but I'm not sure.'

4. Compositionally implementing the Conditional Belief Schema

One project: find new (context-relative) semantic values for sentences, some but not all of which correspond to propositions, and new values for sentential operators including 'if' and 'believes', such that when A and C have semantic values that corresponding to propositions p_1 and p_2 , 'X believes that if A , C ' has the semantic value corresponding to the proposition that X 's conditional credence in p_2 given p_1 is above a contextually determined threshold.

An alternative: adopt a Kratzer-inspired picture, where:

- Not only modals like 'must' and 'might', but also attitude words like 'believes', 'thinks', and 'confident' take (at least) two sentence-type arguments, not one.
- The 'if' clause and 'main' clause deliver these arguments separately, i.e. $\llbracket \text{Jim believes that if it's raining the streets are wet} \rrbracket$ is $\llbracket \text{believes} \rrbracket(\llbracket \text{it's raining} \rrbracket)(\llbracket \text{the streets are wet} \rrbracket)(\llbracket \text{Jim} \rrbracket)$. If the 'if' clause is absent it's treated as if it was some tautology.
- In a bare conditional, there's a hidden operator that also takes two sentence-type arguments, which does not (or does not have to) occur in a conditional embedded under a modal or attitude verb.
- When P and Q have semantic values corresponding to propositions p_1 and p_2 , $\llbracket \text{believes} \rrbracket(P)(Q)(x)$ has a semantic value corresponding to the proposition that x 's conditional credence in p_2 given p_1 is above a contextually determined threshold.

All of this is consistent both with an orthodox proposition-based semantics for modals including the hidden one, and with expressivist/dynamic alternatives. But Seth of course would like to implement it in a domain-semantic friendly way, like this:

$\llbracket \text{believes} \rrbracket(\llbracket \phi \rrbracket)(\llbracket \psi \rrbracket)(\llbracket A \rrbracket)^{w,s} = 1$ iff $\llbracket A \rrbracket$'s conditional credence at w in $\{w': \llbracket \phi \rrbracket^{w', I(w,A)} = 1\}$ given $\{w': \llbracket \psi \rrbracket^{w', I(w,A)} = 1\}$ is above t .

5. Some worries about the Kratzer-inspired picture.

- (i) It might seem to be wedded looks to require the rather implausible idea that the true *syntactic structure* of 'Jim believes that if P, Q' is $\llbracket \text{Jim} \llbracket \text{believes that} \llbracket \text{if P} \rrbracket, Q \rrbracket$. But as Chris pointed out to me a couple of weeks ago, we can avoid this using type raising. E.g. we could have something like

'believes': $t \rightarrow t \rightarrow e \rightarrow t = \lambda p q x. x$'s conditional credence in q given p is high.

'if': $t \rightarrow t \rightarrow (t \rightarrow t \rightarrow e \rightarrow t) \rightarrow e \rightarrow t = \lambda p q X y. X p q y$

- (ii) It doesn't deal well with sentences where conditionals combine with other things under an attitude report, as in 'Jim believes that A and if B, C' or 'Jim is pretty confident that if A, B and if C, D' or 'Jim hopes that either B if A or D if C'. It looks like the only way of dealing with these would be to introduce the same hidden modal that arises with unembedded conditionals, which (at least if it works like 'must') will generate bad results: 'Jim is pretty confident that A and if B, C' is not the same as 'Jim is pretty confident that A and it must be that if B, C'.
- (iii) There's a worry about cases where we seem to anaphorically refer to the arguments of 'believe'. E.g. consider this dialogue (from von Stechow):

A: If he didn't tell Harry, he told Tom. B: Probably so. B': That's very unlikely.

We could also have followups like 'I think so', 'I'm fairly confident that that's the case but not as confident as you seem to be', etc.

Even if we aren't wedded to the idea that the semantic value of 'that' (or 'so') has to correspond in some straightforward way to the semantic value of some previous constituent, we still need 'that's the case' to contribute something that makes 'I'm fairly confident that that's the case' true! But what could this be?

6. Background to the second paper: a puzzle about 'might' under 'knows'.

A puzzle about 'might': 'Bill knows that I might be in that café' seems acceptable in lots of settings where 'I might be in that café' would be weird. Is this a counterexample to the validity of the argument from 'Bill knows that P' to P?

- A contextualist reaction: No. 'I might be in that café' has distinct Bill-centric and me-centric readings. The Bill-centric reading is true, and is the default under 'Bill knows that...'; the me-centric reading is false, and is the default reading unembedded.
- A possible expressivist reaction: yes, it's a counterexample. This argument-form is only valid when P is "factual".

- Maybe this isn't so bad.
- This reaction will be endorsed if one extends the semantics for 'suppose' in Yalcin 2007 to 'know' in the obvious way. 'Bill knows that I might be in that café' is a factual sentence true exactly when some world where I am in that café is compatible with every proposition Bill knows: it's equivalent to 'Bill doesn't know that I am not in that café'.
- A big problem for this view: since only true propositions can be known, this makes the inference from 'P' to 'Bill knows that it might be that P' valid.
- A somewhat kludgy alternative: 'Bill knows that I might be in that café' is not a factual sentence; it is equivalent to 'I might be in that café, and Bill doesn't know [or believe?] that I am not in that café'.

7. How this extends to conditionals

Yalcin's third puzzle: 'Bill knows that if I'm not in this café I'm in the one across the street' seems acceptable in lots of settings where 'If I'm not in this café I'm in the one across the street' would be weird. Is this a counterexample to the validity of the argument from 'Bill knows that P' to P?

- A possible reaction here: the weirdness of the unembedded conditional is not a matter of its being false, or not believed, but just a matter of presuppositions.
- Compare: we might want to say that 'Every N Vs' logically implies 'Every N that is A Vs', and appeal to presupposition to account for the oddity of the inference from 'Every elephant has a trunk' to 'Every elephant in the room has a trunk'.

But there are puzzles just about presuppositions. Let's talk about presupposition as a relation between sentence-types, analogous to logical implication, and closed under it (if P presupposes Q and Q logically implies R, P presupposes R. Here are some appealing generalizations:

Generalization 1: 'If P, Q' presupposes 'It might be that P'

Generalization 2: 'X knows that P' presupposes P

Generalization 3: If P presupposes Q, 'X knows that P' presupposes Q

Generalization 4: If P presupposes Q, 'X knows that P' presupposes 'X knows that Q'.

- Consider: 'Jim doesn't know that my dog is a Labrador' — 'Hey wait a minute, no-one told me Jim knew you had a dog'.

Yalcin's first puzzle: generalizations 2 predicts that 'Bill knows that if we're not at this café we're at the one across the street' presupposes 'If we're not at this café we're at the

one across the street', but this seems wrong since the former can be fine to assert when the latter would be weird.

Yalcin's second puzzle: generalizations 1 and 3 predict that 'Bill knows that if we're not at this café we're at the one across the street' presupposes 'We might be at the café across the street', but this seems false since the former can be fine to assert when the latter would be weird.

- A possible reaction: Generalizations 2 and 3 fail in general, though they hold for factual P.
- But note that we could keep Generalization 4: maybe 'Bill knows that if we're not at this café we're at the one across the street' does presuppose 'Bill knows that we might be at the café across the street'.